Calculus and Linear Algebra

Finite Limits: Meaning, Definition and Theorems

Lecture 13

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Lecture outline

- What a limit describes: behaviour near a point, not necessarily at the point.
- Left-hand and right-hand limits.
- Finite limits at a finite point.
- Rigorous ε–δ definition.
- Uniqueness, sign/order properties and the squeeze theorem.
- Algebra of finite limits.
- First calculations: direct evaluation, factorization and rationalization.

Main goal

Understand the concept before learning the computational techniques.

The central question of a limit

Suppose x moves closer and closer to a number a. What happens to $f(x)$?

$$x \to a \qquad \Longrightarrow \qquad f(x) \to ?$$

The answer may exist even when

- $f(a)$ is different from the limiting value;
- $f(a)$ is not defined at all.

A limit is about nearby inputs.

The value at the single point $x = a$ is not what defines the limit.

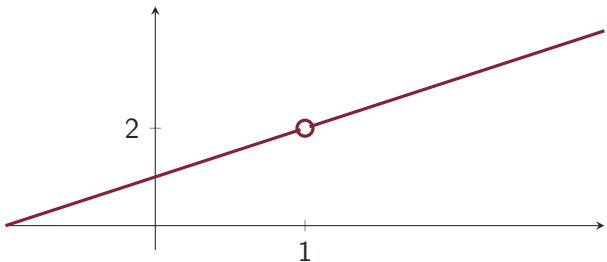
A first example: a hole that does not affect the limit

For $x \neq 1$,

$$f(x) = \frac{x^2 - 1}{x - 1} = x + 1.$$

The original formula is undefined at $x = 1$, but nearby values approach 2.

$$\lim_{x \to 1} \frac{x^2 - 1}{x - 1} = 2.$$



Limit notation

We write

$$\boxed{\lim_{x \to a} f(x) = L}$$

to mean that $f(x)$ can be made arbitrarily close to L by taking x sufficiently close to a, with $x \neq a$. Read it as:

> *"the limit of $f(x)$ as x tends to a is L."*

Here a is the **point approached** and L is the **limiting value**.

Approaching from the left and from the right

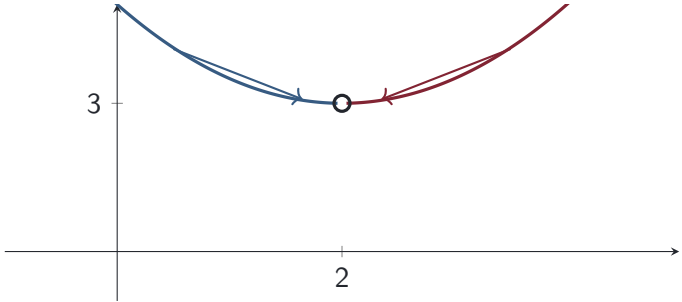
There are two possible directions of approach:

$$\boxed{\lim_{x \to a^-} f(x)} \quad \text{and} \quad \boxed{\lim_{x \to a^+} f(x)}.$$

- $x \to a^-$: values smaller than a approach a.
- $x \to a^+$: values larger than a approach a.

These are called the **left-hand** and **right-hand** limits.

When a two-sided limit exists

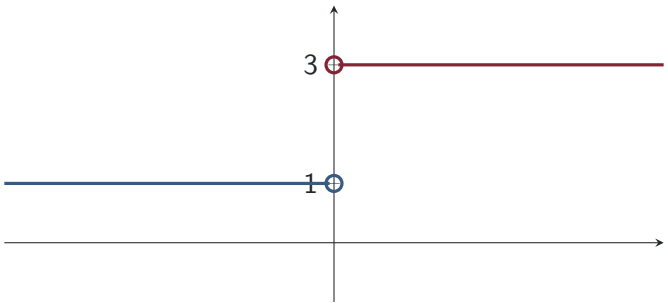


Both sides approach the same value:

$$\lim_{x \to 2^-} f(x) = 3, \qquad \lim_{x \to 2^+} f(x) = 3.$$

Therefore $\lim_{x \to 2} f(x) = 3$.

When the two-sided limit does not exist



$$\lim_{x \to 0^-} f(x) = 1, \qquad \lim_{x \to 0^+} f(x) = 3.$$

Since the one-sided limits are different, the two-sided limit does not exist.

One-sided criterion

Theorem A finite two-sided limit exists exactly when both one-sided limits exist and are equal:

$$\boxed{\lim_{x \to a} f(x) = L \iff \lim_{x \to a^-} f(x) = L \text{ and } \lim_{x \to a^+} f(x) = L.}$$

Practical consequence

For piecewise functions, absolute values and possible jumps, always inspect the two sides separately.

Finite limit at a finite point: neighbourhood idea

The statement

$$\lim_{x \to a} f(x) = L$$

means that when x is confined to a sufficiently small neighbourhood of a, the corresponding values $f(x)$ are confined to an arbitrarily small neighbourhood of L.
Input closeness:

$$0 < |x - a| < \delta.$$

Output closeness:

$$|f(x) - L| < \varepsilon.$$

The symbols δ and ε measure horizontal and vertical tolerance.

Rigorous definition of a finite limit

Definition Let f be defined in a punctured neighbourhood of a. We say

$$\boxed{\lim_{x \to a} f(x) = L}$$

if

$$\boxed{\forall \varepsilon > 0 \ \exists \delta > 0 \text{ such that } 0 < |x - a| < \delta \Rightarrow |f(x) - L| < \varepsilon.}$$

Order matters

The tolerance ε is chosen first. We must then find a suitable δ that guarantees the requested output accuracy.

What the quantifiers are saying

$$\forall \varepsilon > 0$$

means: no matter how narrow a vertical band around L is requested.

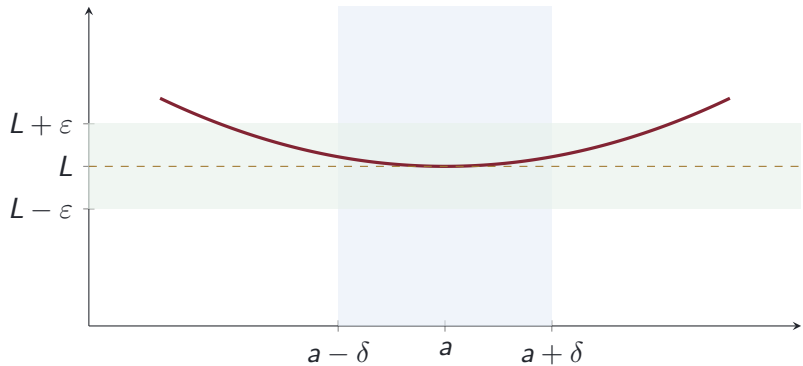
$$\exists \delta > 0$$

means: we can choose a horizontal window around a.

$$0 < |x - a| < \delta \Rightarrow |f(x) - L| < \varepsilon$$

means: every graph point with x inside that punctured horizontal window lies inside the requested vertical band.

The ε–δ picture



The δ-window must be chosen so that the graph near a remains inside the ε-band around L.

A complete ε–δ verification

Verify

$$\lim_{x \to 3} (2x + 1) = 7.$$

We need

$$|(2x + 1) - 7| < \varepsilon.$$

But

$$|(2x + 1) - 7| = |2x - 6| = 2|x - 3|.$$

Therefore it is enough to impose

$$|x - 3| < \frac{\varepsilon}{2}.$$

Choose

$$\boxed{\delta = \frac{\varepsilon}{2}.}$$

Then $0 < |x - 3| < \delta$ implies $|2x + 1 - 7| < \varepsilon$.

The limit ignores the value at the point

Consider a function defined by

$$f(x) = \begin{cases} x + 1, & x \neq 1, \\ 10, & x = 1. \end{cases}$$

Then

$$f(1) = 10,$$

but

$$\boxed{\lim_{x \to 1} f(x) = 2.}$$

Changing the value of a function at a single point does not change its limit at that point.

The limit can exist even if the function is undefined

Again,

$$f(x) = \frac{x^2 - 1}{x - 1}$$

is undefined at $x = 1$, yet

$$\lim_{x \to 1} f(x) = 2.$$

Key distinction

$\boxed{f(a)}$ and $\boxed{\lim_{x \to a} f(x)}$

are different mathematical objects. They coincide only under an additional property: continuity.

Uniqueness of the limit

Theorem If a finite limit exists, it is unique. That is, it is impossible to have simultaneously

$$\lim_{x \to a} f(x) = L_1 \qquad \text{and} \qquad \lim_{x \to a} f(x) = L_2$$

with $L_1 \neq L_2$.

Why this matters

A function cannot approach two different finite values at the same point from both sides.

Proof of uniqueness

Assume, for contradiction, that $L_1 \neq L_2$. Choose

$$\varepsilon = \frac{|L_1 - L_2|}{3} > 0.$$

For x sufficiently close to a, both limit assumptions would imply

$$|f(x) - L_1| < \varepsilon, \qquad |f(x) - L_2| < \varepsilon.$$

By the triangle inequality,

$$|L_1 - L_2| \leq |L_1 - f(x)| + |f(x) - L_2| < 2\varepsilon = \frac{2}{3}|L_1 - L_2|,$$

a contradiction. Therefore $L_1 = L_2$.

Permanence of sign

Theorem If

$$\lim_{x \to a} f(x) = L > 0,$$

then $f(x) > 0$ for all x sufficiently close to a (possibly excluding a itself). Similarly, if $L < 0$, then $f(x) < 0$ near a.

Interpretation

A function converging to a strictly positive number cannot keep taking negative values arbitrarily close to the limiting point.

Order theorem

Theorem Suppose

$$f(x) \leq g(x)$$

for all x sufficiently close to a, and suppose both finite limits exist. Then

$$\boxed{\lim_{x \to a} f(x) \leq \lim_{x \to a} g(x).}$$

This theorem allows inequalities between functions to pass to their limits.

Squeeze theorem

Theorem Suppose that near a

$$g(x) \leq f(x) \leq h(x),$$

and

$$\lim_{x \to a} g(x) = \lim_{x \to a} h(x) = L.$$

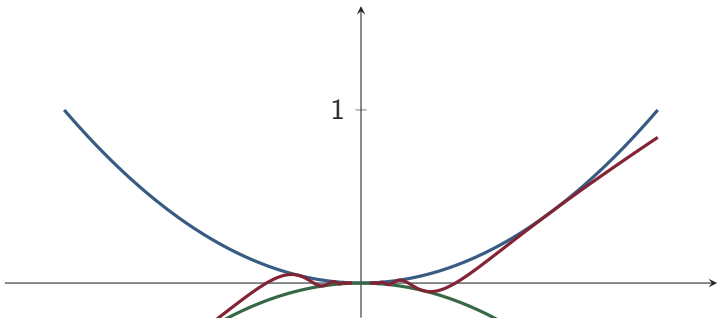
Then

$$\boxed{\lim_{x \to a} f(x) = L.}$$

Picture

If a function is trapped between two functions that both approach the same value, it has nowhere else to go.

Squeeze theorem: graphical intuition



Since

$$-x^2 \leq x^2 \sin(1/x) \leq x^2$$

and both bounds tend to 0, the middle function also tends to 0.

Algebra of finite limits

Suppose

$$\lim_{x \to a} f(x) = L, \qquad \lim_{x \to a} g(x) = M.$$

Then

$$\lim_{x \to a} [f(x) + g(x)] = L + M,$$

$$\lim_{x \to a} [f(x) - g(x)] = L - M,$$

$$\lim_{x \to a} [cf(x)] = cL,$$

$$\lim_{x \to a} [f(x)g(x)] = LM.$$

These rules reduce many limits to simpler ones.

Quotients, powers and roots

Under the same assumptions,

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \frac{L}{M} \qquad \text{provided } M \neq 0.$$

For positive integers n,

$$\lim_{x \to a} [f(x)]^n = L^n.$$

When the root is defined near a,

$$\lim_{x \to a} \sqrt[n]{f(x)} = \sqrt[n]{L}.$$

Warning

The quotient theorem cannot be used when the denominator limit is zero. That situation requires separate analysis.

Direct evaluation: when it works

For polynomials, substitution gives the limit directly:

$$\lim_{x \to 2} (x^3 - 4x + 1) = 8 - 8 + 1 = 1.$$

For rational functions, direct evaluation works when the denominator does not tend to zero:

$$\lim_{x \to 1} \frac{x^2 + 3}{x + 2} = \frac{4}{3}.$$

This follows from the algebraic limit theorems.

The symbol 0/0 is not an answer

Consider

$$\lim_{x \to 1} \frac{x^2 - 1}{x - 1}.$$

Direct substitution produces

$$\frac{0}{0}.$$

This is an **indeterminate form**, not the value of the limit. It tells us only that direct substitution is insufficient. We must transform the expression while preserving it for $x \neq 1$.

Factorization removes a removable obstacle

$$\lim_{x\to 1}\frac{x^2-1}{x-1}=\lim_{x\to 1}\frac{(x-1)(x+1)}{x-1}.$$

For $x\neq 1$,

$$\frac{(x-1)(x+1)}{x-1}=x+1.$$

Therefore

$$\boxed{\lim_{x\to 1}\frac{x^2-1}{x-1}=2.}$$

The cancellation is legitimate for a limit because we only need values **near** 1, not at 1 itself.

Another factorization example

Compute

$$\lim_{x \to 2} \frac{x^2 - 5x + 6}{x - 2}.$$

Factor:

$$x^2 - 5x + 6 = (x - 2)(x - 3).$$

For $x \neq 2$,

$$\frac{x^2 - 5x + 6}{x - 2} = x - 3.$$

Hence

$$\boxed{\lim_{x \to 2} \frac{x^2 - 5x + 6}{x - 2} = -1.}$$

Rationalization

Compute

$$\lim_{x\to 0} \frac{\sqrt{1+x}-1}{x}.$$

Multiply numerator and denominator by the conjugate:

$$\frac{\sqrt{1+x}-1}{x} \cdot \frac{\sqrt{1+x}+1}{\sqrt{1+x}+1} = \frac{1}{\sqrt{1+x}+1}.$$

Therefore

$$\boxed{\lim_{x\to 0} \frac{\sqrt{1+x}-1}{x} = \frac{1}{2}.}$$

A limit from an inequality

Compute

$$\lim_{x \to 0} x \sin\left(\frac{1}{x}\right).$$

Since

$$-1 \leq \sin(1/x) \leq 1,$$

we have

$$-|x| \leq x \sin(1/x) \leq |x|.$$

Both bounds tend to 0. By the squeeze theorem,

$$\boxed{\lim_{x \to 0} x \sin(1/x) = 0.}$$

Reading limits from a graph

When a graph is given, do not calculate. Ask:

1. What value does the graph approach from the left?
2. What value does it approach from the right?
3. Are the two values equal?
4. What is $f(a)$? Is it relevant to the limit?

Keep the questions separate

The graph may have a hole, a jump or an isolated point. Each affects $f(a)$ and the limit differently.

Guided exercise: one-sided limits

For

$$f(x) = \begin{cases} 2x + 1, & x < 1, \\ 4 - x, & x \geq 1, \end{cases}$$

find

$$\lim_{x \to 1^-} f(x), \qquad \lim_{x \to 1^+} f(x), \qquad \lim_{x \to 1} f(x), \qquad f(1).$$

Guided exercise: solution

From the left,

$$\lim_{x \to 1^-} (2x + 1) = 3.$$

From the right,

$$\lim_{x \to 1^+} (4 - x) = 3.$$

Therefore

$$\boxed{\lim_{x \to 1} f(x) = 3.}$$

Also

$$f(1) = 4 - 1 = 3.$$

In this example the function value and the limit agree; later this will be called continuity at $x = 1$.

Independent practice

Compute or decide whether the limit exists.

1. $\lim_{x \to 2} (3x^2 - x + 1)$.
2. $\lim_{x \to -1} \dfrac{x^2 + 2x + 1}{x + 1}$.
3. $\lim_{x \to 0} \dfrac{\sqrt{4 + x} - 2}{x}$.
4. $\lim_{x \to 0} |x|$.
5. If $f(x) = 1$ for $x < 0$ and $f(x) = 2$ for $x > 0$, find the one-sided and two-sided limits at 0.

Independent practice: answers

1. 11.
2. Factor $(x+1)^2$: the limit is 0.
3. Rationalize: the limit is $\frac{1}{4}$.
4. 0.
5. Left-hand limit 1, right-hand limit 2; the two-sided limit does not exist.

What we know about finite limits

We can now distinguish

- the value $f(a)$;
- the left-hand limit;
- the right-hand limit;
- the two-sided finite limit.

We also have a rigorous definition and the main structural theorems.

Next lecture

The limiting value need not be finite, and the input itself need not approach a finite point. We next study $\pm\infty$ rigorously.